

Model Performance Report

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System Overview

A snapshot of the current Piezo.AI platform state, including datasets loaded, models trained, predictions made, and database utilization.

Metric	Value
Datasets	3
Total Material Rows	73
Trained Models	3
Predictions Made	6
Training Jobs	1
Database Size	8.1 MB

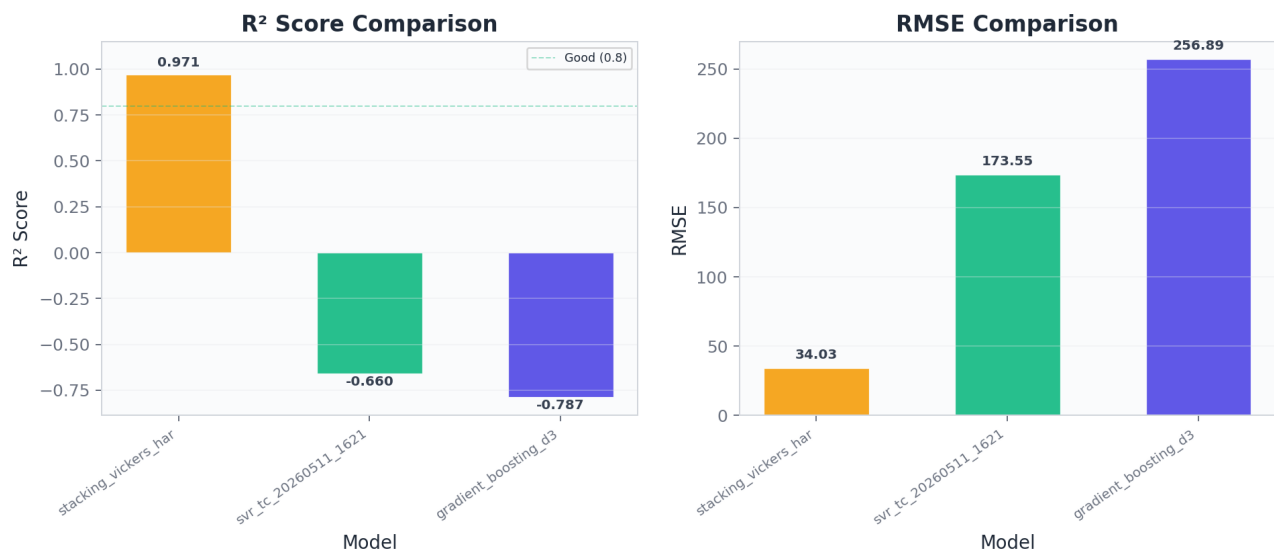
Trained Models Summary

Overview of all trained ML models with their target property, algorithm, R² goodness-of-fit score, RMSE prediction error, and training/test split sizes.

Name	Target	Algorithm	R ²	RMSE	Train	Test
stacking_vickers_hardness_20	HV	stacking	0.9712	34.03	24	6
svr_tc_20260511_162120	Tc	svr	-0.6600	173.55	24	6
gradient_boosting_d33_202605	d33	gradient_boosting	-0.7868	256.89	24	6

R² and RMSE Comparison

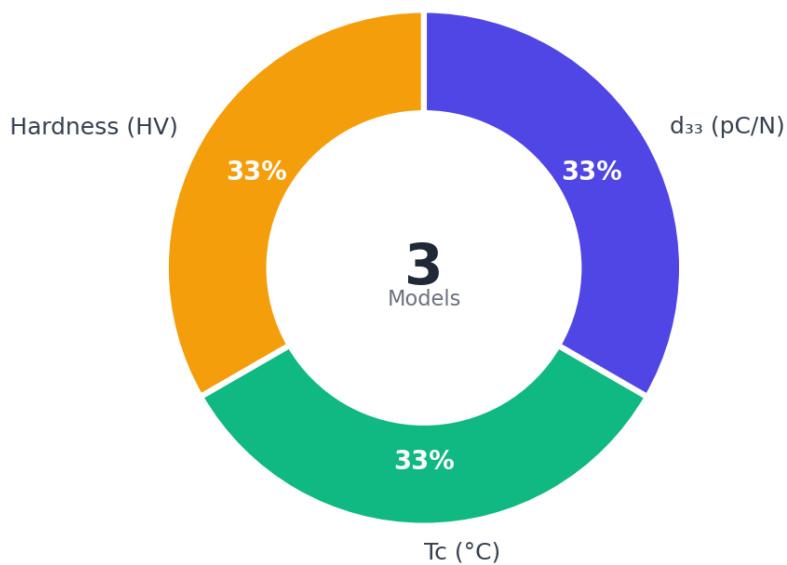
Side-by-side comparison of R² (coefficient of determination) and RMSE (root mean square error) across all trained models. Higher R² and lower RMSE indicate better model accuracy and reliability for material property predictions.



Model Target Distribution

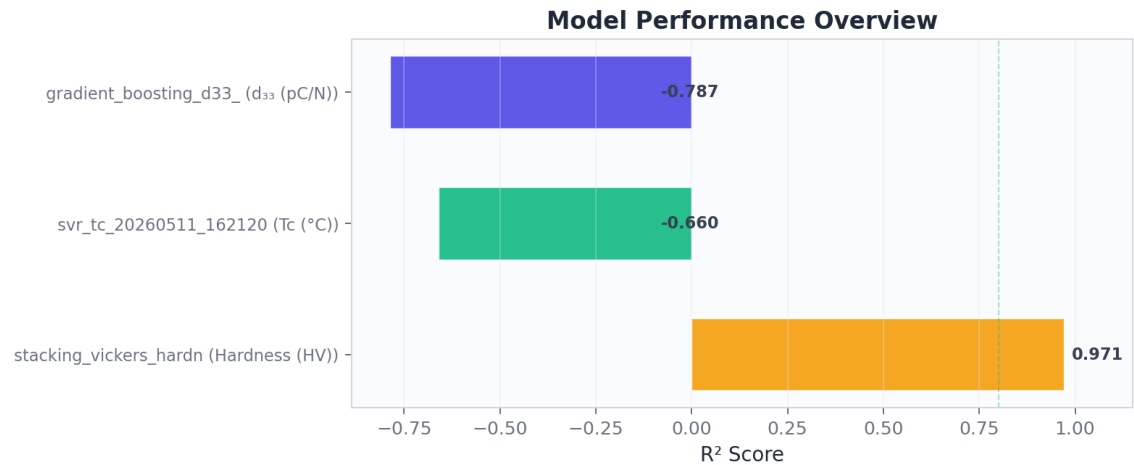
Distribution of trained models across target properties (d_{33} , T_c , Hardness). Shows the proportion of models dedicated to predicting each material property, indicating research focus and coverage balance.

Model Target Distribution



Performance Overview

Visual comparison of model performance via R^2 scores. The dashed line at 0.8 represents the threshold for reliable predictive quality. Models exceeding this threshold are considered suitable for guiding experimental material synthesis.



AI Performance Analysis

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The current performance of the Piezo.AI platform demonstrates a significant disparity between the prediction of mechanical and functional properties. The Vickers hardness model is excellent, achieving an R^2 of 0.9712, which indicates a high degree of reliability for screening the mechanical durability of new ceramic compositions. In contrast, the models for Curie temperature and the piezoelectric charge coefficient are statistically invalid, as evidenced by their negative R^2 values. These negative metrics signify that the models perform worse than a simple average of the training data, making them currently unsuitable for guiding experimental material discovery.

The failure of the d_{33} and T_c models is particularly critical because these parameters define the operational limits and efficiency of piezoelectric devices. An RMSE of 256.89 for d_{33} is nearly as large as the expected value for many standard ceramics, suggesting that the gradient boosting algorithm is unable to map the complex, non-linear relationships inherent in piezoelectricity. Functional properties in these systems are often driven by sensitive phenomena such as morphotropic phase boundaries and domain wall mobility, which require more sophisticated feature engineering or significantly more data to resolve than what is currently provided.

The primary limitation across all models is the insufficient dataset size, with only 24 training samples available. While mechanical properties like hardness may follow relatively simple composition-based trends that can be captured with small samples, functional piezoelectric properties are highly dependent on processing history and crystalline symmetry. To improve model accuracy, it is essential to expand the training set to at least several hundred entries. Additionally, the inclusion of physics-informed descriptors, such as the Goldschmidt tolerance factor, ionic radii ratios, and specific sintering conditions, would likely help the models better distinguish between high-performance and low-performance compositions.

AI Material Insight

The predicted properties of the $\text{PbCaNbO}_3\text{-ZnSrTa}_2$ system suggest a versatile range of applications in the fields of sensing and energy harvesting. With d_{33} values ranging from 162.3 to 175.5 pC/N, both the bulk ceramic and composite forms are well suited for ultrasonic transducers, acoustic sensors, and vibration based energy harvesters. The Curie temperature of 183.0 degrees Celsius ensures stable operation for standard industrial and consumer electronics, though it remains below the threshold required for high temperature aerospace or automotive engine environments. The bulk ceramic variant is particularly notable for its high hardness of 530.1 HV, making it an excellent candidate for ultrasonic motors or structural health monitoring sensors where mechanical robustness is essential. In contrast, the lower hardness of the composite variant suggests its utility in flexible electronics or medical imaging applications where acoustic impedance matching with soft tissue is a priority.

When compared to established piezoelectric materials, this system demonstrates a piezoelectric response that exceeds standard barium titanate and is comparable to many potassium sodium niobate compositions. However, it does not reach the high piezoelectric coefficients typically associated with soft lead zirconate titanate, which often exceeds 300 pC/N. The composition property relationships reveal that while the Curie temperature remains constant regardless of the material type, the mechanical properties

are highly sensitive to the microstructure. The transition from a bulk ceramic to a composite results in a significant reduction in hardness but a slight increase in the d_{33} value. This suggests that the composite structure may allow for greater dipole mobility or reduced mechanical clamping, thereby enhancing the piezoelectric charge constant at the expense of structural rigidity. This trade off is a critical consideration for engineers selecting between these two forms for specific device architectures.

Prediction Insights

Predicted material properties for selected formulas, including d_{33} piezoelectric coefficient, Curie temperature (T_c), Vickers hardness, and the top suggested use case based on the combination of predicted properties.

Formula	d_{33}	T_c	HV	Status
PbCaNbO3-ZnSrTa2	175.5	183.0	127.8	Success